

ZZSC9020-Data Science Project (H525 Online)

Assessment 1 Part A: Group Project Plan

Enhancing Electricity Demand Forecasting in NSW Using Synthetic Data Under Extreme Temperature Conditions

Team Members:

Curtis Lauro (z5415940),

Maria Kim (z3115676),

Mae Li (z5571254),

Anqi Zou (z5353786),

Jeff Eker (z3445077)

Table of Contents

1. Introduction and Motivation	3
2. Brief Literature Review	3
3. Methods, Software and Data Description	4
4. Activities and Schedule.....	4
4.1 Team Roles and Responsibilities	4
4.2 Project Activities.....	5
4.3 Project Timeline.....	5
References.....	6

1. Introduction and Motivation

Extreme weather events, such as heatwaves, can cause sharp surges in electricity demand as households and businesses increase air-conditioning usage. These spikes place significant stress on the power grid, making accurate short-term demand forecasting critical for maintaining system stability and avoiding blackouts.

While electricity demand forecasting has been widely studied, most models are trained on normal operating conditions. Techniques such as Random Forests [1], XGBoost [2], and LSTM [3, 4] rely on historical demand, temperature, and time-of-day features. However, these models often struggle to generalise under rare, high-demand conditions due to limited examples in historical datasets. This challenge stems from the fact that extreme weather days are relatively rare in historical datasets, leading to a scarcity of representative training examples. As a result, a model trained on “normal” days may significantly underpredict electricity demand during an outlier event.

To address this issue, we propose generating synthetic electricity demand profiles that mimic the patterns observed during extreme temperature conditions. By incorporating these synthetic examples into the training process, we aim to expose the model to a broader range of rare but realistic scenarios, thereby improving its robustness and accuracy when predicting demand under stress.

2. Brief Literature Review

Electricity demand forecasting has increasingly shifted from traditional statistical models [6, 7] toward advanced machine learning models such as Random Forests [1], XGBoost [2], and deep neural networks like LSTMs [3, 4], which effectively capture temporal dependencies and nonlinear patterns in historical data. While these models show strong generalisation capabilities under normal conditions, their accuracy tends to degrade under extreme conditions such as heatwaves or cold snaps, largely due to the scarcity of such events in historical datasets [5].

To mitigate this, recent work has explored synthetic data generation to augment datasets, particularly for power consumption time series, a category emphasised in recent surveys [13]. Techniques such as Gaussian Mixture Models (GMMs) [14, 15] and deep generative models including Generative Adversarial Networks (GANs) [9, 10, 11, 12, 13] and Conditional Variational Autoencoders (cVAEs) [8] have been used to produce realistic load profiles.

Despite these efforts, very few studies generate synthetic electricity demand specifically conditioned on temperature extremes, or evaluate the impact on real-world forecast accuracy during such events. Our project fills this gap by generating synthetic electricity demand under extreme temperature scenarios, using GMM, GAN, and VAE-based approaches, and assessing their benefit in downstream forecasting models.

3. Methods, Software and Data Description

To address data scarcity under extreme weather conditions, we will generate synthetic electricity demand profiles using *GMM*, *GAN*, and *VAE*. These techniques have been successfully applied in augmenting time-series data and improving model robustness, particularly under rare scenarios.

Our dataset comprises **total electricity demand** and **forecast demand** in half-hourly intervals for New South Wales, sourced from the Market Management System database. **Air temperature** data is sourced from the Bankstown Airport weather station via the Australian Data Archive for Meteorology.

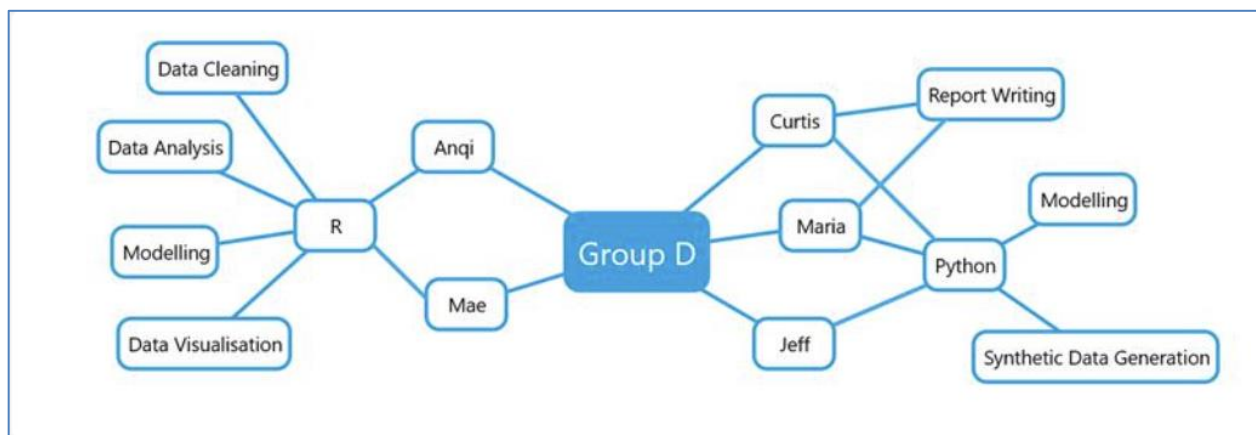
For electric demand forecasting, we will train baseline models using *Random Forest* (robust, interpretable), *XGBoost* (strong with limited data and noise) and *LSTM* (capture temporal dependencies), comparing performance with and without synthetic data. Model evaluation will focus on accuracy and robustness during known extreme temperature events.

We will use R, Python, and PyTorch for exploratory data analysis (EDA) and modelling, leveraging Google Colab with GPU acceleration. Key libraries include *scikit-learn*, *xgboost*, *statsmodels*, *matplotlib/seaborn*, and *PyTorch*.

4. Activities and Schedule

4.1 Team Roles and Responsibilities

Curtis Lauro serves as the *Project Lead*, overseeing coordination, progress tracking, and milestone delivery. Maria Kim, as *Research Lead*, is responsible for the literature review, problem definition, and proposing methodological directions. Mae Li, Anqi Zou, and Jeff Eker support the project as *Data Analysts*, handling data cleaning, model implementation, evaluation, and visualisation tasks. While these are the primary roles, all members will actively contribute to model implementation, experimentation, reporting, and presentation preparation. A visual summary of team responsibilities is provided in the mind map below.



4.2 Project Activities

The project is divided into five major work stages:

Stage	Activity Description
1. Data Preparation & EDA	Load, clean and explore the given datasets; identify extreme weather events and align data for model input.
2. Model Development	Train baseline models on the original dataset and evaluate their performance under both normal and extreme conditions.
3. Synthetic Data Generation	Generate synthetic demand profiles conditioned on extreme temperatures.
4. Model Training with Augmented Data	Retrain the same models with synthetic data and compare their performance to the original baseline.
5. Evaluation & Reporting	Analyse improvements in accuracy and robustness, and prepare the final report with visualisations.

4.3 Project Timeline

The project timeline and task activity mapping are visualised in the Gantt chart below.

Task	Week 1	Week 2	Week 3	Week 4	Week 5	Week 6
1. Project Plan + Literature Review	✓	✓				
2. Data Preprocessing + EDA		✓	✓			
3. Baseline Model Development + Evaluation			✓	✓		
4. Synthetic Data Generation			✓	✓		
5. Model Retraining using synthetic data + Evaluation				✓	✓	
6. Report Writing		✓	✓	✓	✓	✓
7. Presentation Preparation					✓	✓

References

1. G. Dudek, "A comprehensive study of Random Forest for short-term load forecasting," *Energies*, vol. 15, no. 20, p. 7547, Oct. 2022.
2. N. Shabbir, R. Ahmadianhangar, A. Rosin, M. Jawad, J. Kilter, and J. Martins, "XGBoost based short-term electrical load forecasting considering trends & periodicity in historical data," in *Proc. 2023 IEEE Int. Conf. on Energy Technologies for Future Grids (ETFG)*, Sydney, Australia, Dec. 2023, pp. 1–6.
3. N. Al Khafaf, M. Jalili, and P. Sokolowski, "Application of deep learning Long Short-Term Memory in energy demand forecasting," arXiv preprint arXiv:1903.11941, 2019.
4. M. Bouktif, A. Fiaz, A. Ouni, and M. Serhani, "Optimal deep learning LSTM model for electric load forecasting using feature selection and genetic algorithm: Comparison with machine learning approaches," *Energies*, vol. 11, no. 7, p. 1636, 2018.
5. H. Liu, Z. Ma, L. Yang, T. Zhou, R. Xia, Y. Wang, Q. Wen, and L. Sun, "SaDI: A self-adaptive decomposed interpretable framework for electric load forecasting under extreme events," arXiv preprint arXiv:2306.08299, 2023.
6. Baharudin, Z.; Kamel, N. Autoregressive Method in Short Term Load Forecast. In Proceedings of the 2008 IEEE 2nd International Power and Energy Conference, Johor Bahru, Malaysia, 1–3 December 2008; pp. 1603–1608.
7. Jin, X.; Dong, Y.; Wu, J.; Wang, J. An Improved Combined Forecasting Method for Electric Power Load Based on Autoregressive Integrated Moving Average Model. In Proceedings of the 2010 International Conference of Information Science and Management Engineering, Xi'an, China, 7 August 2010; Volume 2, pp. 476–480.
8. F. Cheng, M. Chen, R. Tang, and J. Wang, "A novel deep generative modeling-based data augmentation strategy for improving short-term building energy predictions," *Building Simulation*, vol. 15, no. 2, pp. 197–211, 2022.
9. J. Barta, A. J. van der Meer, and W. T. de Boer, "Synthetic electricity demand data generation using conditional generative adversarial networks," *Energy AI*, vol. 8, p. 100158, 2022.
10. N. Sailor, M. Hoerling, R. Kumar, and T. Moreau, "Generation of synthetic benchmark electrical load profiles using generalized weather-driven techniques," *Energy*, vol. 78, pp. 87–98, 2015.
11. B. Yilmaz and R. Korn, "Synthetic demand data generation for individual electricity consumers: Generative Adversarial Networks (GANs)," *Energy and AI*, vol. 10, p. 100161, 2023.
12. S. Asre and A. Anwar, "Synthetic Energy Data Generation Using Time Variant Generative Adversarial Network," *Electronics*, vol. 11, no. 3, p. 355, Jan. 2022.
13. M. Turowski, B. Heidrich, L. Weingärtner, L. Springer, K. Phipps, B. Schäfer, R. Mikut, and V. Hagenmeyer, "Generating synthetic energy time series: A review," *Renewable and Sustainable Energy Reviews*, vol. 195, p. 114842, Aug. 2024.

14. R. Singh, B. C. Pal, and R. Jabr, "Statistical representation of distribution system loads using Gaussian mixture model," *IEEE Transactions on Power Systems*, vol. 25, no. 1, pp. 29–37, 2010.
15. Y. Li, B. Hu, T. Niu, S. Gao, J. Yan, K. Xie, and Z. Ren, "GMM-HMM-based medium- and long-term multi-wind farm correlated power output time series generation method," *IEEE Access*, vol. 9, pp. 90255–90267, 2021.